

Predicting U.S. Bank Distress

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PROJECT SUMMARY

A model that flags which U.S. banks are heading for undercapitalization, using the quarterly reports every bank already files with the FDIC plus national economic data. The recommendation is that supervisors rank the review queue with a machine learning model instead of ranking based on their capital ratio alone. Modeling is finished. This week was the final deck.

MILESTONES

1. Literature review of bank failure and existing early-warning models (**DONE**)
2. Collect the data (FDIC API, 1984 to present) (**DONE**)
3. Understand the data through exploratory data analysis (**DONE**)
4. Prepare the data for machine learning (**DONE**)
5. Train and compare four candidate models (**DONE**)
6. Redefine the target on a Tier 1 capital basis (**DONE**)
7. Build the executive visuals (**DONE**)
8. Storyboard the final presentation (**DONE**)
9. Build the final deck (**DONE**)
10. Rehearse it and prepare the defense (**DONE**)

LAST WEEK'S "TO DO"

From the Week 6 report: build the ten slides against the storyboard headlines and rehearse the five-minute version. (**DONE**)

THIS WEEK'S PROGRESS

- Built the final deck, eleven slides. Two are new since the storyboard: one that introduces the model, and a limits slide.
- Checked the deck against the Week 7 time budget. It maps one to one.
- Named the audience: FDIC supervisory staff who run offsite monitoring and set examination schedules. Not executives.

- Analyzed the model results more thoroughly. Compared with ranking on capital, the model finds 30 more troubled banks and misses 6 that capital catches, a net 24.
- Wrote the decision log.
- Drafted the LLM reflection.

DECISION LOG

1 How undercapitalization is measured

Week 5. The regulation puts a bank in the worst capital category when its tangible equity falls to 2% of assets or below, and it measures that equity as Tier 1 capital. Mine used book equity. I rebuilt the label on Tier 1 and re-ran every model against it. The alternative was to leave it alone: book equity is simpler, it was already built, and it gave me 563 cases to measure on instead of 160. I would only have kept it if the regulation defined it that way, and it does not.

2 Where to split train from test

Week 4. Train through 2015, drop all of 2016, test from 2017. A random split was the obvious alternative and it is wrong here: shuffling rows puts 2008 and 2023 on both sides, so the model gets tested on the crisis it trained on. A time split with no gap is wrong too, because a training row from the end of 2015 is labeled on what happens through the end of 2016, so that label would already describe the test period. The gap is four quarters wide because that is exactly how far ahead the label looks. The same horizon costs me the end of the data as well, 5,637 of the 166,754 rows I test on, which have no outcome yet.

3 Whether to feed the model economic data

The model reads 98 measures from each bank's own filing. I also had 11 national economic series, things like interest rates and unemployment. Week 4, I kept them. On the 2017 to 2025 test years the model is slightly better without them, so in calm years they are not earning their place. I kept them because when I retrain on older data and test on the 2007 to 2011 crisis instead, they help a lot. A model a supervisor keeps on the shelf has to work in the bad years. If it were only ever going to be used in calm ones, I would drop them.

4 Which method does the ranking

Week 4, I chose gradient boosting, which builds a few hundred small decision trees and adds them together. I expected it to win: my Week 3 exploration found several warning signs declining together for about two years before trouble, and gradient boosting is the method here that learns combined effects, where falling capital matters more because bad loans are rising at the same time. I tested three alternatives on exactly the same rows: logistic regression, and two kinds of neural network. Gradient boosting ranked troubled banks best, the capital ratio came second, and all three alternatives came in behind it. I wanted logistic regression to win, because it gives you one number per measure that you can read out loud and gradient boosting gives you nothing you can explain that way. I would have taken it if the gap had been small enough to be chance.

5 What to compare the model against

Week 4, I compared it with ranking banks by their capital ratio, which is the measure the regulation itself uses to define the event I predict. The stronger claim would have been beating SCOR, the FDIC's own early-warning model, and I took SCOR out of my results table the same day. SCOR answers a different question: whether a supervisor downgrades the bank over the next one to three years, not whether it falls below a capital minimum within one. Putting them head to head would have compared two different questions and flattered whichever model drew the easier one. If SCOR predicted the same event, I would have benchmarked against it.

LLM REFLECTION (DRAFT)

Where it was wrong

Often. It was reliable when I gave it specific instructions, but it made things up several times. I never asked for the things it made up, it produced them on its own and with confidence. I had to double check every claim.

In Week 1 it proposed arbitrary choices, such as what to use as the target and which time windows to select. In Week 4 it proposed dropping columns I had selected from the literature review, then reversed the moment I asked. This week it told me my model cannot see a deposit run because it did not predict SVB, when in fact my model reads 20 funding measures including deposits, and the reason it did not predict SVB is that SVB's regulatory capital never fell, so the event my model predicts never happened.

Where I overrode it, and what I checked myself

I went back to 12 C.F.R. § 324.403 myself and found my own label measured the wrong equity, so I rebuilt it and re-ran every model. I dropped the NBER recession flag once I realised it is dated after the fact and could not have been known in the quarter it marks. I kept the columns I had chosen from the literature review when it wanted them dropped. I ran three alternative methods on the same rows rather than accept a recommendation. And I took SCOR out of the comparison after checking what event it actually predicts.

Honest stance

I leaned on it heavily for the modelling, but I also used it a lot for brainstorming, which probably cost me more time than it saved. The good habit I developed is fact-checking every important claim I was not sure about, so as not to spread misinformation.

ISSUES & DISCUSSION

Covering every requirement while explaining everything plainly runs over the time limit. I tried to find a balance, assuming the audience has some grounding in data science as well as in banking.

NEXT WEEK'S "TO DO"

- Record the full timed run and post it with a self-assessment.

- Finish the LLM collaboration log.

RESOURCES

Repository: <https://github.com/oussamaennaciri/us-bank-undercapitalization-early-warning>

Near-final deck: [presentations/final-draft.pdf](#), eleven slides.

Note on LLM use: This document was typeset in LaTeX using Claude Code, which handled the formatting, styling code, and wording polish. All content, research, and decisions are my own, and every suggested edit was reviewed and approved by me.